Forensic Microscopy

Lab 8: Natural Fibers

Aritra Bhattacharya
Graduate teaching assistant
ecoevoaritra23@tamu.edu

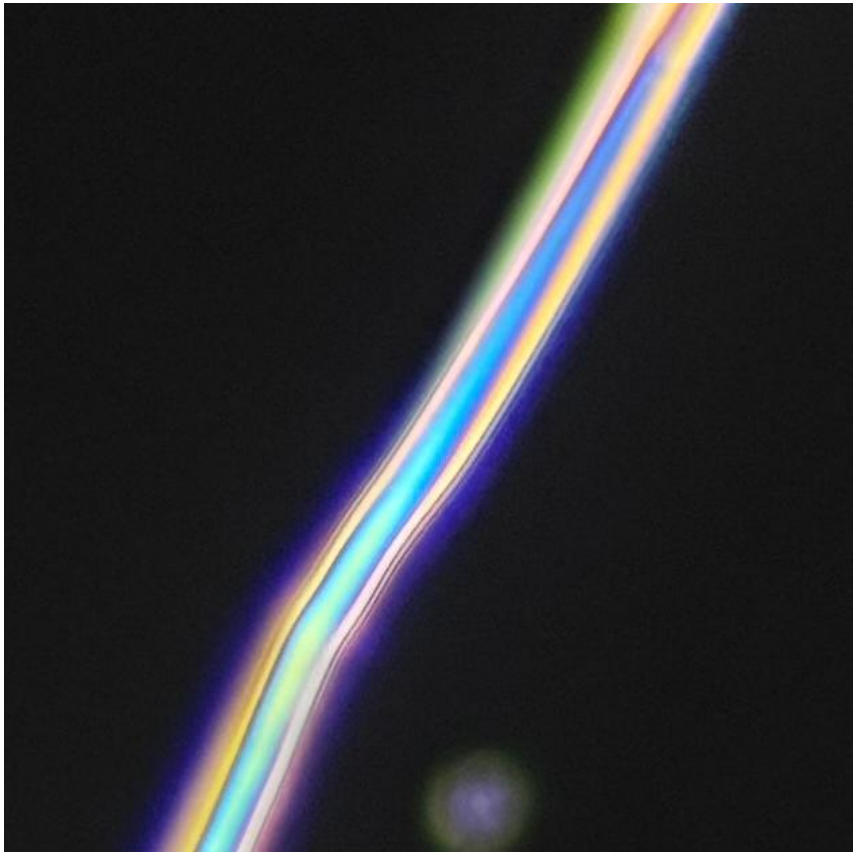
Jackie Maddox
Undergraduate teaching assistant
jac.r.maddox@tamu.edu

Where can you find the slides?

- https://aritra-evolves.github.io/FIVS215.html
- Canvas

Agenda

- Coming Up/Reminders
- Types of Fibers
- Viewing
- Open Lab



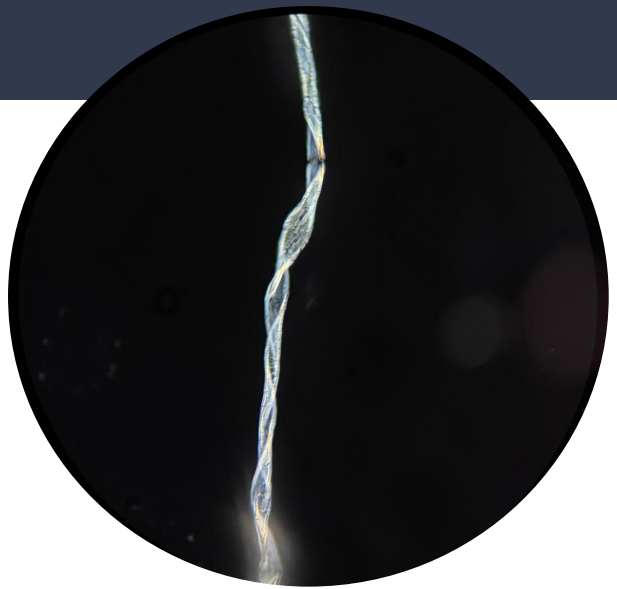
Seed Fibers

Cotton

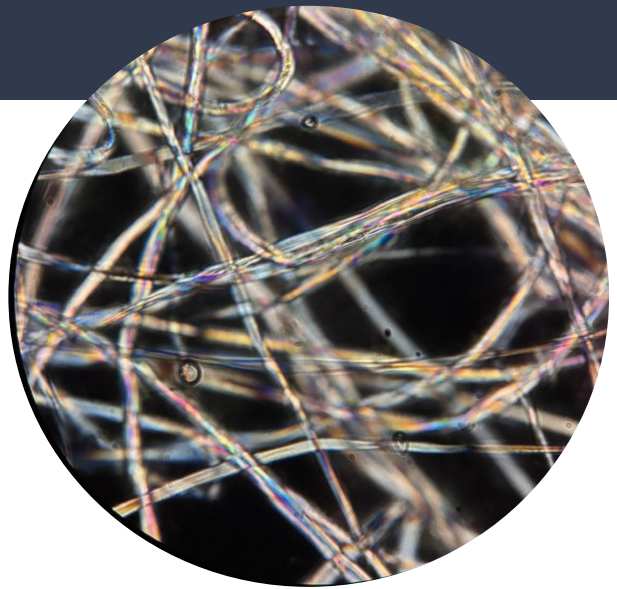
- Retardation Color
- Incomplete Extinction
- Undulating
- Lumen
- Flattened

Kapok

- Used in Floatation Devices
- Not as Common Anymore



Cotton



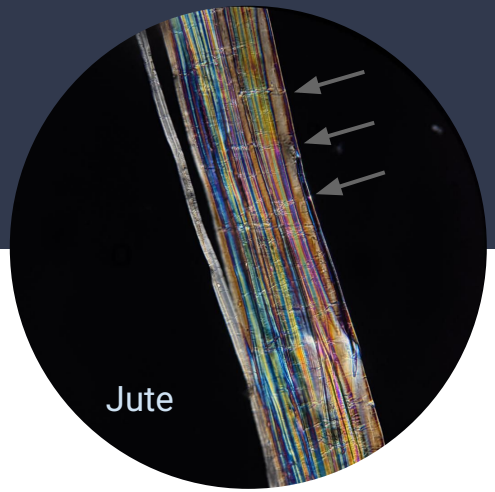
M. Cotton
(fibers are more straight)

Cotton fibers of different colors look similar under microscope

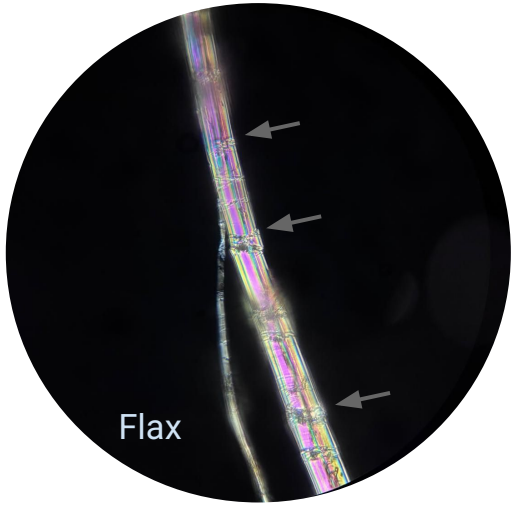
Bast Fibers

- Straighter
- Dislocations
- Less Common in Clothing

- Flax (Linen)
- Hemp
- Jute
- Sisal



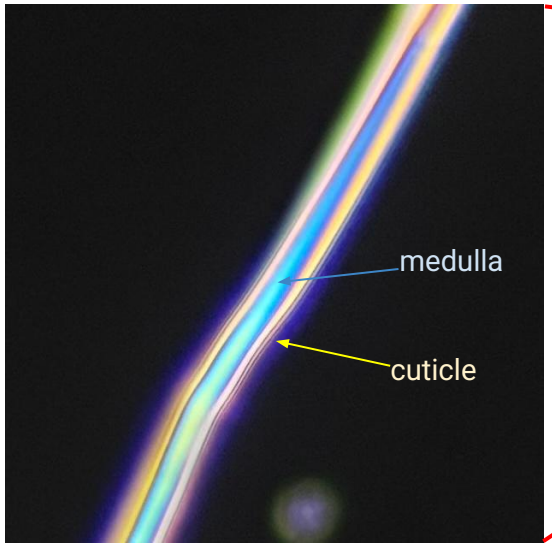


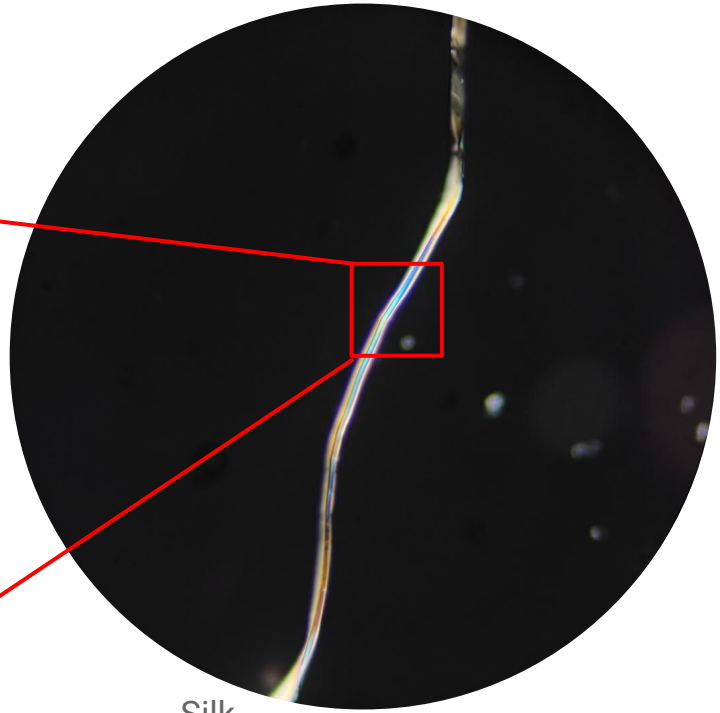




Animal Fibers

- Medulla and Cuticle
- Wool & Silk Most Common
- Used in Clothing





Silk

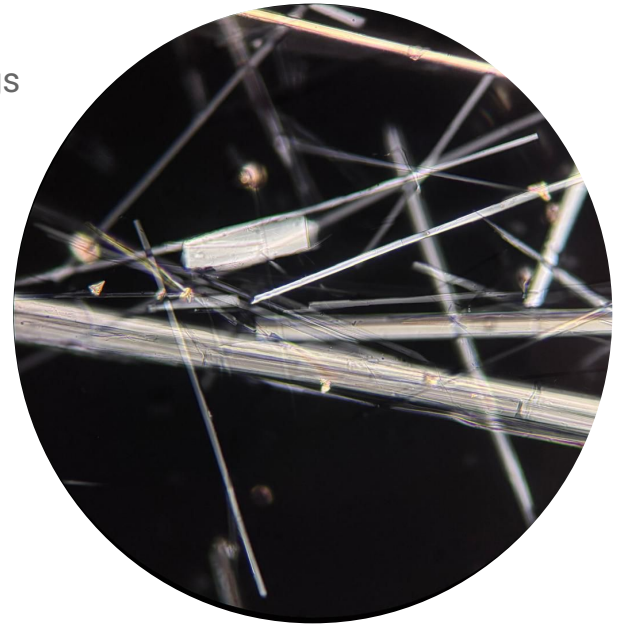
Mineral Fibers & Fiberglass

Asbestos

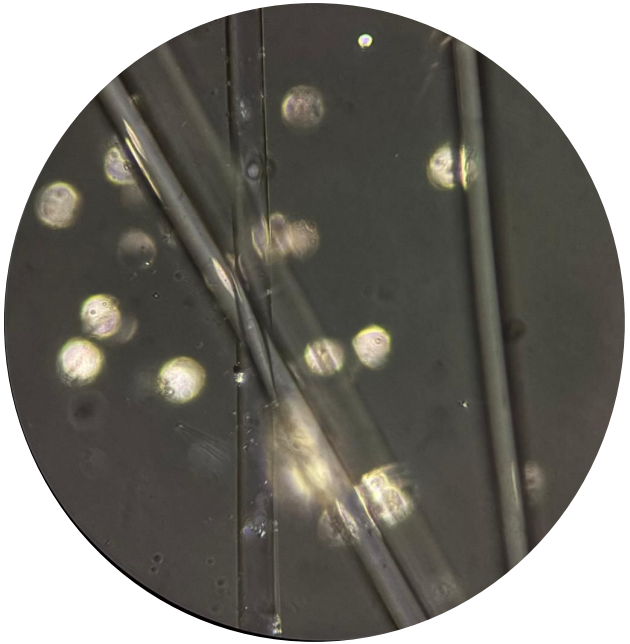
- Very common in older buildings
- Chrysotile & Amosite

Fiberglass

- Isotropic
- Commercial



Asbestos



Fiber glass

Practice

1. Grab a Slide from the Natural Box
2. Set Up Kohler
3. Cross Polars to View Colors

Note

1. What You See
2. Isotropic vs. Anisotropic

Next Week

- More Time on This OR Open Lab

Cotton
Jute
Flax
Hemp
Silk
Fiberglass
Asbestos

Bring back the slides in it's original place

See you next week!